Chapter 9 — Big-data tools, lakes, and warehouses

Streaming needs a landing place

Learning objectives

- By the end of this part, learners should be able to list integration, storage, quality

Collection challenges at big-data scale

- Big data arrives from social, sensors, databases
- Traditional relational stores struggle with volume and variety
- Quality drifts under velocity

Tools: Flume and NiFi

- Often into real-time analytics paths
- Apache NiFi automates data flow between systems
- These tools address movement and integration when custom scripts do not scale

Data lakes for raw, mixed sources

- A data lake is a centralized repository that stores vast raw data in native formats
- Lakes suit text, video, and social feeds when analysts need flexible access later
- Weak schema discipline

Example 9.26 — Retail Data Lake for Mixed Sources

- Example 9.26 — hands-on module
- Example 9.26 places customer interaction logs, social content
- Analysts later apply machine learning to uncover behavior trends from that mixed corpus
- Explore the chapter example module
- View files: `modules/chapter9/example26/`

Data warehouses for structured BI

- A data warehouse stores cleaned
- Tables support business intelligence tools and historical analysis
- Warehouses excel at governed reporting but are less suited to the full variety and

Example 9.27 — Financial Data Warehouse for Structured BI

- Example 9.27 — hands-on module
- Example 9.27 shows a financial institution storing transactions, balances
- Structured BI needs consistency that lakes alone may not provide
- Explore the chapter example module
- View files: `modules/chapter9/example27/`

Takeaways

- Big-data collection faces integration, storage, quality
- Lakes land mixed raw sources for flexible analytics
- Retail lakes and financial warehouses illustrate complementary storage patterns

Next

- Complete the quiz for this part
- Emerging trends that prepare later storage and modeling chapters